



Liquidity commonality in the cryptocurrency market

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ABSTRACT

Motivated by the unique transaction cost structure of the cryptocurrency (CC) market¹, this study investigates the phenomenon of liquidity commonality across a sample of 53 CCs. The study employs the google search volume index (GSVI) measure to capture the retail investor's attention towards the CC market. Using the quantile regression method, we document the liquidity dynamics of CCs that is contrasting to other asset classes, and is ascribed to its unique transaction cost structure. In view of the relatively high liquidity commonality levels found in the CC market, this paper sounds a note of caution to retail investors on episodic non-availability of liquidity in CCs.

KEYWORDS

Liquidity commonality; Cryptocurrency; Principal component analysis; Quantile regression; Transaction costs

JEL CLASSIFICATION

G12; G14; C33

I. Introduction

Cryptocurrency (CC, hereafter) market – an emerging asset class – has witnessed significant and rising investor attention globally. Since the inception of Bitcoin in 2009 (Nakamoto 2008), the CC market has reached an aggregate capitalization of USD 215.58 billion² – with a CAGR of approximately 65% over the last six years (2013–19). Cryptocurrencies are not regulated by state authorities such as central banks, and therefore, lack the conventional sovereign support. CCs operate on a peer-to-peer online payment system between private parties, without involving regulatory authorities and financial intermediaries (Corbet et al. 2019). CC markets are generally characterized by (a) continuous trading on highly fragmented³ and unregulated markets, (b) speculative trader base, and (c) dearth of institutional investors. Unlike traditional markets, cryptocurrencies rely on an open-source software that creates these currencies, and records and verifies the transactions (mining process, Easley, O'Hara, and Basu 2019). The stock of cryptocurrencies is increasing at a diminishing rate that is expected to converge to zero by the year 2140.

In the last 3–5 years, a number of studies have examined various important strands of CC literature, including security, price efficiency, volatility, price

clustering and bubbles, long memory and heteroscedasticity, and regulatory aspects including their role as a medium of exchange (Corbet et al. 2019; Smith and Kumar 2018). However, the liquidity dynamics of CCs is relatively less-examined. To the best of our knowledge, few studies examine the liquidity of cryptocurrencies, and most of them focus on Bitcoin (Brauneis and Mestel 2018; Dyhrberg, Foley, and Svec 2018; Fry 2018; Koutmos 2018; Wei 2018). These currencies are not driven by the economic fundamentals (macroeconomic, industry, and firm-specific factors) that explain conventional asset classes' liquidity dynamics. Therefore, the liquidity characteristics of these CCs are expected to differ considerably from those of a traditional security. The conventional demand and supply side determinants of liquidity (e.g. trading activity of dealers and designated market makers, the role of institutional investors) become less effective in the context of CC markets (Amihud 2002; Amihud & Mendelson, 1986; Chordia, Roll, and Subrahmanyam 2000; Easley, O'Hara, and Basu 2019; Ho and Stoll 1983; Karolyi, Lee, and Van Dijk 2012; Koch, Ruenzi, and Starks 2016; Stoll 1978; Tripathi, Dixit, and Vipul 2019). This makes the liquidity dynamics in CC markets unique, and an interesting research proposition. More importantly, unlike any other asset class, increasing volumes (beyond certain threshold levels)

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²As per the <https://coinmarketcap.com/> (accessed on 21-11-2019).

³Bitcoin alone is traded on more than 350 markets across the world, in various currencies (Wei 2018).

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drive-up the transaction fees in CC markets (Easley, O'Hara, and Basu 2019). Because CC users are required to pay transaction fees to get their transactions posted on blockchain. As the transaction volumes rise and transaction processing capacities are fully utilized, the waiting times for users increase substantially. These users compete for priority to reduce the waiting time, and in turn, drive-up the transaction fees. The transaction processing priority is dynamically auctioned by the miners with auction price being the transaction fee. Higher fees secure higher priority and less waiting time.

These unique features of the liquidity in CC markets motivate us to examine the presence of a marketwide systematic factor that has been widely acknowledged to affect the liquidity of traditional securities; i.e. the phenomenon of liquidity commonality (Chordia, Roll, and Subrahmanyam 2000). The literature on liquidity commonality has covered a diverse set of markets. These include equity markets (cash and derivatives, developed and emerging markets), fixed income securities markets, currency markets, and commodity markets. A summary of some of the most important studies on commonality, covering these diverse markets, is provided in Table A1 of Appendix A. These studies document the evidence of liquidity commonality and its adverse implications for investors. The literature on commonality supports the argument that the benefits of portfolio diversification and various other trading strategies disappear when the transaction costs are accounted for, especially when liquidity commonality is high. However, the liquidity dynamics of CCs remains less well understood, and in particular, the area of liquidity commonality of CC markets has not been explored (Marshall, Nguyen, and Visaltanachoti 2019). Hence, the results from this study are particularly important to investors that are primarily driven towards CCs by their portfolio concerns. This study provides evidence on the phenomenon of liquidity commonality in the context of the CC market, using 53 cryptocurrencies.

Another strand of literature examines the role of investor attention on security prices and liquidity (Barber and Odean 2007; Ding and Hou 2015; Hirshleifer, Lim, and Teoh 2011). The literature suggests that increasing investor attention decreases information asymmetry and improves

the liquidity of securities. More recent literature has employed Google Search Volume Index (GSVI) as a measure of investor attention (Da, Engelberg, and Gao 2011, 2014; Ding and Hou 2015). We contribute to this literature by adding a novel and less explored asset class, that is CCs.

The study makes three important contributions to the literature. It is one of the first studies that examines the phenomenon of liquidity commonality in the CC market, employing 53 CCs. The study also investigates the effect of investor attention on liquidity dynamics of CCs, using the daily GSVI measure. Finally, the study examines commonality across the conditional quantiles of liquidity, employing the quantile regression framework (Koenker and Bassett 1978; Koenker and Hallock 2001). The findings of this study suggest a significant presence of commonality that is higher than other asset classes. The results show that investor attention has a positive relation with volume as well as transaction costs. These findings are in contrast to those pertaining to other asset classes, and may be ascribed to the unique transaction cost structure that is integral to the CC market. That is, higher volumes (beyond certain threshold levels) drive-up the transaction fees and vice-versa. We further show that this property of the CC market attenuates the escalating commonality levels around the extreme quantiles of liquidity.

The remaining study is organized into four sections. Section 2 describes the data, sample, and variables, and Section 3 discusses the methodology. Section 4 presents the empirical findings, and Section 5 concludes.

II. Data, sample, and variables

Daily price and volume data of cryptocurrencies are provided by the Coinmarketcap website (www.coinmarketcap.com). Coinmarketcap aggregates the trading activity of more than 2,200 cryptocurrencies across about 8,900 exchanges. It computes the volume-weighted average price of cryptocurrency, converted into USD. We select the 53 cryptocurrencies among the top-500 for which the trading history is available for the period spanning 1 January 2017–31 October 2019. These currencies comprise about 85% of the total cryptocurrency market capitalization (Table 1). Out of these 53

CCs, there are 49 coins and 4 tokens (Tether, Augur, Madsafe, and Golem).⁴ The descriptive statistics of the top-5 currencies are provided in Table 2.

For measuring the liquidity of CCs, we use widely employed proxies of bid-ask spread that can be calculated using the daily price and volume⁵ data. These are Amihud measure (Amihud 2002; 'Amihud'), high-low price measure (Corwin and Schultz 2012; 'CS'), high-low-close range measure (Będowska-Sójka 2018; 'RHL'), high-low range measure (Chung and Zhang 2014; 'HLR'), and volatility over volume measure (Fong, Holden, and Tobek 2018; 'VOV'). The formulae for these measures are as follows:

$$(a) \text{Amihud}_t = \frac{|r_t|}{\log(\text{volume}_t)}$$

$$(b) \text{CS}_t = \frac{2(e^{\alpha_t} - 1)}{1 + e^{\alpha_t}};$$

where,

$$\alpha_t = \frac{\sqrt{2\beta_t} - \sqrt{\beta_t}}{3 - 2\sqrt{2}} \sqrt{\frac{\gamma_t}{3 - 2\sqrt{2}}}, \beta_t = \left[\ln\left(\frac{H_t}{L_t}\right) \right]^2 + \left[\ln\left(\frac{H_t + 1}{L_t + 1}\right) \right]^2 \text{ and } \gamma_t = \ln\left(\frac{\max\{H_t, H_{t+1}\}}{\min\{L_t, L_{t+1}\}}\right)$$

$$(c) \text{RHL}_t = \frac{H_t - L_t}{C_t \frac{H_t - L_t}{H_t + L_t}}$$

$$(d) \text{HLR}_t = \frac{H_t - L_t}{0.5(H_t + L_t)}$$

$$(e) \text{VOV}_t = \frac{\ln\left(\frac{H_t}{L_t}\right)}{\sqrt{\log(\text{volume}_t)}}$$

Here, H_t , L_t , C_t , and volume_t are high, low, closing prices, and volume on day t ; r_t is the price return on day t based on closing prices. We also employ the daily Google Search Volume Index (GSVI) for these CCs as a proxy of investor attention (Da, Engelberg, and Gao 2011, 2014). Given the dominance of retail investors in CC market, the GSVI measure of investor attention appears to be a reasonable choice. We use daily GSVIs corresponding to full currency names (e.g. Bitcoin) and

their symbols (e.g. BTC) for all the CCs in our sample set. Prior studies, however, have relied on weekly and monthly search volume index data to proxy retail investor attention.

The descriptive statistics for these liquidity measures (for the top-5 currencies) are provided in Table 3. The average-correlations (averaged over the 53 CC's) across these measures are reported in Table 4. Furthermore, average-correlations of these measures with GSVIs (search terms: 'Cryptocurrency' and 'Bitcoin') are also shown. GSVIs exhibit a positive correlation with volume measure and various spread measures. The positive correlation with spread measures are contradictory to those observed in cases of other asset classes. It suggests that high investor attention is related to increasing volumes along with increasing transaction costs. Since the keyword 'Cryptocurrency' is a broad keyword associated with the asset class, we may expect it to exhibit a higher correlation than the keyword 'Bitcoin', which is related to an individual (though major) currency.

III. Methodology

The study follows two approaches to measure commonality: (a) Principal component and canonical correlation analyses approach of Hasbrouck and Seppi (2001), using the liquidity and GSVI measures in normalized form.⁶ (b) Market model of CRS (2000) shown in Equation (1), which is further augmented [Equation (2)] with control variables [GSVI for CC full name/symbol, and return and volatility⁷ of individual CC and the market]. Here, $DL_{i,t} = \ln\left(\frac{L_{i,t}}{L_{i,t-1}}\right)$ denotes the daily changes in the liquidity of the i -th Cryptocurrency (CC) from day $t-1$ to day t , and is the liquidity variable of interest (Chordia, Roll, and Subrahmanyam 2000; Karolyi, Lee, and Van Dijk 2012; Marshall, Nguyen, and Visaltanachoti 2013). $L_{i,t}$ is the liquidity measure

⁴While the term 'token' is often employed interchangeably with 'coin', there are important differences (Howell, Niessner, and Yermack 2018; Tasca 2019).

Tokens (e.g. Tether) are CCs that do not have their own blockchain networks. Unlike coins, tokens do not act as a store of value, as they derive value from some other asset (similar to derivatives). That is, they digitally represent some other asset (such as gold, U.S. Dollar). Moreover, unlike coins, tokens are not mined. For example, Tether operates on the Ethereum network. It digitally represents the U.S. Dollar, and therefore, derives its value from the U.S. Dollar. The inclusion of tokens in the study is supported on two grounds. (a) Tokens operate on the same blockchain network as coins (predominantly Ethereum and Binance Coin) and trade on the CC markets. (b) The investor profile remains similar for both the CCs, i.e. tokens and coins. Therefore, if observed, the presence of commonality can be largely ascribed to the CC market microstructure and similar investor profile.

⁵The volume measure is computed as the natural log of the daily dollar trading volume.

⁶The normalization procedure involves scaling the difference between the observation and time-series mean by the standard deviation of the series.

⁷Return-squares are employed to proxy volatility. The results remain qualitatively similar with the application of conditional volatility obtained from GARCH models (not reported here for brevity).

Table 1. Details of cryptocurrencies (sourced from Coinmarketcap.com on 21–11-2019).

No.	Name	Symbol	Market Cap (USD)	Market share	Price (USD)
1	Bitcoin	BTC	138,008,986,483	65.14%	7,641.67
2	Ethereum	ETH	17,506,660,541	8.26%	161.14
3	Ripple	XRP	10,474,509,984	4.94%	0.24
4	Tether	USDT	4,127,454,673	1.95%	1.00
5	Litecoin	LTC	3,208,410,841	1.51%	50.35
6	Stellar	XLM	1,235,614,975	0.58%	0.06
7	Monero	XMR	940,885,373	0.44%	54.33
8	NEO	NEO	763,365,780	0.36%	10.82
9	Dash	DASH	560,241,051	0.26%	61.13
10	Ethereum Classic	ETC	478,498,117	0.23%	4.15
11	NEM	XEM	343,130,203	0.16%	0.04
12	Dogecoin	DOGE	292,695,461	0.14%	0.00
13	Zcash	ZEC	246,707,888	0.12%	31.23
14	Decred	DCR	201,713,753	0.10%	18.84
15	Augur	REP	126,978,131	0.06%	11.54
16	Lisk	LSK	84,941,091	0.04%	0.70
17	DigiByte	DGB	84,359,289	0.04%	0.01
18	Siacoin	SC	69,134,218	0.03%	0.00
19	Waves	WAVES	65,549,958	0.03%	0.65
20	Verge	XVG	61,428,551	0.03%	0.00
21	Bytecoin	BCN	57,914,953	0.03%	0.00
22	MaidSafeCoin	MAID	50,708,178	0.02%	0.11
23	Ardour	ARDR	47,248,803	0.02%	0.05
24	Steem	STEEM	43,748,776	0.02%	0.12
25	Golem	GNT	39,180,447	0.02%	0.04
26	Zcoin	XZC	36,293,136	0.02%	4.10
27	EDC Blockchain	EDC	35,962,451	0.02%	0.01
28	Stratis	STRAT	29,070,509	0.01%	0.29
29	Factom	FCT	25,913,483	0.01%	2.71
30	DigixDAO	DGD	25,610,602	0.01%	12.81
31	ReddCoin	RDD	21,956,553	0.01%	0.00
32	Obyte	GBYTE	14,385,233	0.01%	19.81
33	Groestlcoin	GRS	13,905,346	0.01%	0.19
34	Nexus	NXS	13,697,147	0.01%	0.21
35	Vertcoin	VTC	12,540,412	0.01%	0.24
36	Syscoin	SYS	12,195,744	0.01%	0.02
37	PIVX	PIVX	11,780,288	0.01%	0.21
38	Nxt	NXT	11,115,121	0.01%	0.01
39	Unobtainium	UNO	9,490,384	0.00%	47.15
40	FirstBlood	1ST	9,054,553	0.00%	0.11
41	Einsteinium	EMC2	8,963,029	0.00%	0.04
42	bitCNY	BITCNY	7,450,027	0.00%	0.14
43	Incnet	INCNT	6,248,470	0.00%	0.14
44	Namecoin	NMC	5,864,712	0.00%	0.40
45	Peercoin	PPC	5,306,171	0.00%	0.21
46	NavCoin	NAV	5,267,648	0.00%	0.08
47	Burst	BURST	5,049,627	0.00%	0.00
48	SaluS	SLS	5,028,641	0.00%	4.97
49	SingularDTV	SINGLS	4,814,879	0.00%	0.01
50	Blocknet	BLOCK	4,788,907	0.00%	0.76
51	MintCoin	MINT	4,239,529	0.00%	0.00
52	DigitalNote	XDN	3,937,616	0.00%	0.00
53	BitBay	BAY	3,243,506	0.00%	0.00

for the i -th CC at time t . The cross-sectional average of $DL_{i,t}$ across all the sample CCs represents the

market liquidity variable, $DL_{M,t} \cdot DSVI_t =$

$\ln \left[\frac{GSVI_{i,t}}{GSVI_{i,t-1}} \right]$ is the measure of investor attention, constructed using daily GSVIs. $r_{i,t}$, $r_{M,t}$, $\sigma_{i,t}$, and $\sigma_{M,t}$ are the returns and standard deviations of individual CCs and the market (at aggregate level).⁸

$$DL_{i,t} = \alpha_i + \beta_1 DL_{M,t} + \beta_2 DL_{M,t+1} + \beta_3 DL_{M,t-1} + \varepsilon_{i,t} \quad (1)$$

$$DL_{i,t} = \alpha_i + \beta_1 DL_{M,t} + \beta_2 DL_{M,t+1} + \beta_3 DL_{M,t-1} + \beta_4 DSVI_{i,t} + \beta_5 \sigma_{i,t}^2 + \beta_6 r_{i,t} + \beta_7 \sigma_{M,t}^2 + \beta_8 r_{M,t} + \varepsilon_{i,t} \quad (2)$$

Quantile regression approach to the CRS (2000) model

The quantile regression approach is particularly relevant in the modelling of conditional quantiles (Koenker and Hallock 2001; Koenker and Xiao 2006). First, the quantile approach is well suited to handle nonlinear relationships that are expected to emerge across different liquidity regimes (Brunnermeier and Pedersen 2008). The approach also offers robustness to estimations around the extreme quantiles of distribution. This is so because security prices often exhibit issues such as heteroscedasticity, skewness, and excess kurtosis, particularly around extreme quantiles of distribution.

The models [Equations (1) and (2)] are reestimated with the quantile regression approach of Koenker (2012). The ' τ -th' conditional quantile function ($0 < \tau < 1$) of the liquidity measure, $DL_{i,t}[Q_{DL_{i,t}}(\tau|DL_{M,t}, DL_{M,t-1}, DL_{M,t+1}, DSVI_{i,t}, \sigma_{i,t}, r_{i,t}, \sigma_{M,t}, r_{M,t})]$, is modelled as the linear function of market liquidity (contemporaneous, lead, and lag), search

⁸Prior literature has suggested the inclusion of lead and lag terms in such market model regressions (Dang et al. 2015; Galariotis and Giouvris 2007; Karolyi, Lee, and Van Dijk 2012; Kempf and Mayston 2008; Marshall, Nguyen, and Visaltanachoti 2013; Moshirian et al. 2017). For example, Chordia, Roll, and Subrahmanyam (2000) state that 'the leads and lags are designed to capture any lagged adjustment in commonality.' Infrequent trading introduces bias in the market model regressions of the kind employed in this study (Dimson 1979). The recorded price and volume information across securities (CCs in this study) may be non-synchronous, and therefore, any market index constructed from such data is an average of temporally ordered values of the securities. Thus, the covariance of less frequently traded securities with the market is underestimated and vice-versa. To account for this bias, Dimson (1979) proposed the aggregated coefficients (AC) method that employs lead, lag, and contemporaneous terms of the market variable. Dimson (1979) argued that infrequently traded securities exhibit higher covariance with the lagged market term. In contrast, heavily traded securities show higher covariance with the leading market term. Therefore, the non-inclusion of lead and lag terms may vitiate the estimation.

Table 2. Descriptive statistics for price returns and unit root tests.

	BTC	ETH	XRP	USDT	LTC
Panel A: Descriptive statistics for return series					
Observations	1,034	1,034	1,034	1,034	1,034
Mean	0.22%	0.30%	0.37%	0.00%	0.25%
Median	0.29%	0.03%	-0.29%	0.00%	-0.08%
Maximum	22.51%	29.01%	102.74%	5.72%	51.03%
Minimum	-20.75%	-31.55%	-61.63%	-4.74%	-39.52%
Std. Dev.	4.39%	5.90%	8.10%	0.72%	6.44%
Skewness	-0.07	0.23	2.81	0.21	1.13
Kurtosis	6.30	6.96	37.65	12.85	12.11
JB [#]	470.66***	685.08***	53,101.42***	4,188.24***	3,794.48***
ARCH(1) ⁺	40.20***	49.39***	87.53***	166.52***	18.23***
ARCH(5)	59.00***	94.02***	97.81***	172.70***	29.29***
Panel B: Unit root test statistics					
ADF [@]	-9.02***	-8.85***	-8.26***	-11.44***	-9.31***
PP ^{&}	-1061.66***	-1128.32***	-1229.57***	-1269.36***	-1064.73***

The table presents descriptive statistics for the top-5 currencies (by market-capitalization) in the sample, over the study period (1 January 2017 to 31 October 2019).

Notes:

[#]Jarque Bera statistic for the significance of skewness and kurtosis differing from a normal distribution

⁺Engle's ARCH Lagrange Multiplier (LM) test

[@]Augmented Dicky-Fuller test

[&]Phillips-Perron test

*, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

volume index, return and standard deviation of individual security and market terms, for different values of τ ($=.05, 0.10, 0.15 \dots .0.95$), as follows.

$$DL_{i,t}[Q_{DL_{i,t}}(\tau | DL_{M,t}, DL_{M,t-1}, DL_{M,t+1})] = \beta_0(\tau) + \beta_1(\tau)DL_{M,t} + \beta_2(\tau)DL_{M,t+1} + \beta_3(\tau)DL_{M,t-1} \quad (3)$$

$$DL_{i,t}[Q_{DL_{i,t}}(\tau | DL_{M,t}, DL_{M,t-1}, DL_{M,t+1}, DSVI_{i,t}, \sigma_{i,t}, r_{i,t}, \sigma_{M,t}, r_{M,t})] = \beta_0(\tau) + \beta_1(\tau)DL_{M,t} + \beta_2(\tau)DL_{M,t+1} + \beta_3(\tau)DL_{M,t-1} + \beta_4(\tau)DSVI_{i,t} + \beta_5(\tau)\sigma_{i,t} + \beta_6(\tau)r_{i,t} + \beta_7(\tau)\sigma_{M,t} + \beta_8(\tau)r_{M,t} \quad (4)$$

The ' $\tau - th$ ' conditional quantile of the stochastic error term is minimized in the quantile regression estimation process. The coefficients $\beta_i(\tau)$'s represent the quantile specific coefficients of the respective variables.

For the implementation of the method, the 'Quantreg R package' developed by Koenker (Koenker 2012; Koenker et al. 2018) is employed. The goodness-of-fit measure for the quantile regression framework (R) is provided in the 'WRTDStidal' package. The function is similar to the conventional R^2 measure. It is computed as one minus the ratio of appropriately weighted sum of absolute deviations from the conditional model to that from the unconditional model (Koenker and Machado 1999).

IV. Empirical results

The following correlation analysis alludes to the presence of commonality across CCs. We compute the values of Pearson correlation coefficient for all possible pairs of CCs, for all the liquidity measures employed in the study. Among these correlations, 88%, 95%, 95%, 96%, 98%, and 97% are positive for volume, Amihud, CS, RHL, HLR, and VoV measures, with average values of 0.37, 0.21, 0.13, 0.21, 0.32, and 0.29, respectively.

To further examine the issue of commonality, first, we perform principal component analysis across CCs, for all the liquidity measures (volume, Amihud, CS, RHL, HLR, and VOV). The results are presented in Panel A (Table 5). The first principal component (PC1) indicates a high level of commonality across CCs, for all the liquidity measures. In particular, the volume measure exhibits a common variation of 43.66%. Next, we examine the commonality in GSVIs for the top-10 and all (53) CCs, separately. The keyword terms are (a) full name of CC (e.g. Bitcoin) and (b) symbol of CC (e.g. BTC). The commonality levels across GSVIs are shown in Panel B (Table 5). The levels of commonality in GSVIs are high, particularly for the top-10 CCs. Finally, we also examine the canonical correlations [CCAs, shown in Panel C (Table 5)] between the liquidity measures (volume, Amihud, CS, RHL, HLR, and VOV) and GSVIs (full name and symbol).

Table 3. Descriptive statistics for daily liquidity measures.

	BTC	ETH	XRP	USDT	LTC
Observations	1,031	1,031	1,031	1,031	1,031
Panel A: Volume measure [log(volume)]					
Mean	22.11	18.76	19.43	20.91	19.85
Median	22.35	19.07	19.86	21.69	19.83
Maximum	24.53	21.27	22.93	24.70	22.66
Minimum	17.92	12.64	12.35	13.43	14.78
Std. Dev.	1.38	1.64	2.01	2.52	1.71
Panel B: Amihud measure (%)					
Mean	0.14%	0.24%	0.23%	0.03%	0.21%
Median	0.09%	0.15%	0.12%	0.02%	0.14%
Maximum	1.00%	2.34%	5.34%	0.27%	2.68%
Minimum	0.00%	0.00%	0.00%	0.00%	0.00%
Std. Dev.	0.14%	0.27%	0.34%	0.03%	0.24%
Panel C: CS measure (%)					
Mean	1.65%	2.59%	2.73%	1.05%	2.32%
Median	1.33%	2.10%	2.15%	0.97%	1.91%
Maximum	18.98%	25.18%	58.20%	7.88%	16.45%
Minimum	0.00%	0.00%	0.00%	0.00%	0.00%
Std. Dev.	1.42%	2.31%	3.44%	0.74%	1.73%
Panel D: RHL measure (%)					
Mean	5.39%	8.35%	8.54%	1.97%	7.78%
Median	4.19%	6.22%	5.58%	1.66%	6.06%
Maximum	27.75%	78.73%	227.23%	20.19%	68.97%
Minimum	0.39%	0.93%	0.92%	0.00%	0.52%
Std. Dev.	4.25%	6.97%	11.47%	1.61%	6.64%
Panel E: HLR measure (%)					
Mean	1.35%	2.08%	2.07%	0.49%	1.93%
Median	1.05%	1.54%	1.40%	0.41%	1.51%
Maximum	7.59%	14.24%	26.71%	4.75%	13.08%
Minimum	0.10%	0.23%	0.23%	0.00%	0.13%
Std. Dev.	1.08%	1.72%	2.26%	0.40%	1.57%
Panel F: VOV measure (%)					
Mean	1.15%	1.92%	1.88%	0.43%	1.73%
Median	0.89%	1.43%	1.28%	0.36%	1.35%
Maximum	6.29%	13.16%	27.16%	4.37%	12.10%
Minimum	0.09%	0.22%	0.21%	0.00%	0.13%
Std. Dev.	0.91%	1.56%	2.07%	0.36%	1.40%

The table provides summary statistics for various daily liquidity measures for the top five CCs (by market-capitalization), for the study period (1 January 2017 to 31 October 2019).

Canonical correlations indicate that liquidity measures and GSVIs share a significant positive common variation. Overall, the results from CCA suggest that

the retail investor attention across CCs is driven by a set of similar factors, and may contribute to the commonality in liquidity.

Next, we follow the market model of CRS (2000) to examine the commonality. The regressions are performed in the fixed-effect panel data framework, and t-statistics are computed using robust standard errors.⁹ The results are shown in Table 6. Panels A and B present the results with and without control variables. For brevity, we show the details only for the contemporaneous market liquidity and DGSVI measures. The contemporaneous market liquidity coefficient is significantly positive for all the liquidity measures and different model specifications. This suggests that, indeed, there is a significant presence of liquidity commonality in the CC market. The results from the pool regression model are qualitatively similar, and therefore, not reported for brevity. These results [shown in Panel C (Table 6)] suggest that the majority of the CCs exhibit a positive and significant coefficient for the contemporaneous market liquidity measures. To examine the robustness of these results, the CRS (2000) model regressions [Equations (1) and (2)] are estimated separately across individual CCs.

The regression coefficients of DGSVI measure (Panel B, Table 6) are consistently positive and significant for most of the liquidity measures, which is contradictory to the results of earlier studies. Previous studies on conventional asset classes suggest that this measure has a negative relation with spread measures. These studies argue that increasing investor attention leads to higher trading volumes and less information asymmetry, and hence, an improvement in liquidity (Adachi,

Table 4. Correlation across liquidity measures and google search volume.

	Volume	Amihud	CS	RHL	HLR	VOV	GSV1	GSV2
Volume	100.00%	−9.56%	−27.90%	−11.87%	−16.88%	−27.15%	8.75%	5.82%
Amihud	−9.56%	100.00%	17.12%	20.28%	22.64%	49.06%	−0.11%	0.36%
CS	−27.90%	17.12%	100.00%	37.25%	52.78%	55.85%	1.21%	3.95%
RHL	−11.87%	20.28%	37.25%	100.00%	84.22%	81.63%	2.35%	5.46%
HLR	−16.88%	22.64%	52.78%	84.22%	100.00%	91.97%	4.12%	8.95%
VOV	−27.15%	49.06%	55.85%	81.63%	91.97%	100.00%	2.24%	6.40%
GSV1	8.75%	−0.11%	1.21%	2.35%	4.12%	2.24%	100.00%	88.63%
GSV2	5.82%	0.36%	3.95%	5.46%	8.95%	6.40%	88.63%	100.00%

The table provides average-correlation across various measures of liquidity for the 53 CCs, over the study period (1 January 2017 to 31 October 2019). Volume measure is computed as the natural log of daily dollar trading volume. GSV1 and GSV2 indicate the daily google search volume index measures for the keywords 'Cryptocurrency' and 'Bitcoin' respectively.

⁹Heteroscedasticity and autocorrelation consistent (HAC) robust standard errors.

Table 5. Commonality in the CCs using Principal Component and Canonical Correlation analyses.

	PC1	PC2	PC3
<i>Panel A: Common variation in liquidity measures explained by the first three principal components (PC's)</i>			
Volume	43.66%	20.90%	4.24%
Amihud	26.44%	3.68%	3.27%
CS	16.91%	3.82%	3.12%
RHL	27.05%	3.84%	3.59%
HLR	36.74%	3.69%	3.15%
VOV	34.69%	4.21%	3.15%
<i>Panel B: Common variation in GSVI measures explained by the first three principal components (PC's)</i>			
Full name of CC			
Top-10 CCs	43.09%	19.70%	13.49%
All 53 CCs	30.78%	9.13%	7.35%
Symbol of CC			
Top-10 CCs	46.19%	17.11%	10.95%
All 53 CCs	24.04%	9.54%	9.26%
<i>Panel C: First two canonical correlations (CVs) between all the liquidity measures and GSVIs</i>			
	Coefficient (%)		Chi-square
Top-10 CCs			
CV1	18.41***		405.29
CV2	6.95***		49.92
All (53) CCs			
CV1	18.38***		1,841.50
CV2	5.01***		125.56

*, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively. Panels A and B report the highest common variation explained by the first three principal components (PCs), for the volume (natural log of dollar trading volume), Amihud, CS, RHL, HLR, and VOV measures, and GSVIs. For GSVIs, we report the PCAs for the top-10 and all the (53) CCs. The GSVI keyword search involves two sets, namely, (a) full currency name (e.g. Bitcoin) and (b) symbol (e.g. BTC). Panel C presents the first two canonical correlations (CCAs: CV1 and CV2) and corresponding Chi-square statistics between the set of two covariates comprising liquidity measures (Volume, Amihud, CS, RHL, HLR, and VOV) and GSVI measures (full name and symbol). The CCAs are presented for the top-10 and all the CCs.

Masuda, and Takeda 2017; Bank, Larch, and Peter 2011; Ding and Hou 2015). Our results indicate that increasing investor attention, though positively related to volume, also has a positive relationship with bid-ask spreads. This contrasting result for the CC market can be ascribed to the unique nature of its transaction cost, which increases with an increase in volumes beyond certain threshold levels.

Quantile approach to commonality

The dynamic transaction costs of CC markets make the commonality in these markets time-varying, and dependent on market liquidity conditions. Hence, the methodologies based on conditional

mean may not fully capture the liquidity dynamics in CCs. To capture this aspect of commonality, we employ the quantile regression methodology. The results from CRS (2000) model [Equation (1)]¹⁰ in a quantile regression framework are summarized in Figure 1 (detailed results are provided in Table B1, Appendix B). We show the t-statistic for the coefficient of contemporaneous market liquidity and goodness-of-fit measure for all the liquidity measures, in Panels A and B.¹¹ These results suggest that the liquidity behaviour for the extreme quantiles (i.e. large positive and negative liquidity changes) differs considerably from that during normal liquidity conditions. Relatively lower commonality is observed for the top and bottom deciles than that for the other quantiles. The plausible reasons for this behaviour are as follows. During the times of extreme positive volume changes in CCs, driven by marketwide systematic shocks, the transaction costs also increase. The excessive increase in transaction costs impedes the trading activity in CC market, and weakens the effect of marketwide systematic factor. Conversely, during the extreme negative volume changes, the transaction costs may come down. This may boost the trading activity in CC market, and in turn, may neutralize the effect of negative marketwide liquidity shocks. This causes the behaviour of liquidity commonality in CC markets to be different from that for conventional asset classes. The levels of commonality are higher than those found for other asset classes. However, in contrast to conventional assets, it decreases for the periods of liquidity shocks.

In summary, we ascribe the higher levels of commonality in CC markets to the following demand and supply related factors. First, the process of mining that governs the supply of cryptocurrencies is similar across all these currencies. Second, even though these currencies are traded across a number of markets, the transaction costs essentially reflect the computational resources spent on mining these currencies and may exhibit uniformity due to similar blockchain management protocols. This leads to an artificial similarity in the microstructure design of all

¹⁰Results from Equation (2) are qualitatively similar, and not reported for brevity.

¹¹The goodness-of-fit measure in quantile regression framework is provided by the 'goodfit' function in the package 'WRDStidal'. In nature, this function is similar to the conventional R^2 measure from regression. The statistic is calculated as one minus the ratio of the appropriately weighted sum of absolute deviations from the conditional model to that from the unconditional model.

Table 6. Liquidity commonality across CCs, using CRS (2000) market model regressions.

Panel A: Results for market model regressions without control variables						
Market Liquidity						
	Number of observations	Coefficient	t-statistic		adjusted.R ² (%)	F-value
Volume	49,958	1.001***	25.33		9.94%	219.94***
Amihud	49,958	0.993***	27.65		19.85%	299.99***
CS	49,958	1.000***	32.60		5.82%	361.56***
RHL	49,958	0.996***	36.63		24.78%	454.33***
HLR	49,958	1.012***	37.33		28.35%	469.91***
VOV	49,958	1.011***	36.44		28.59%	444.85***
Panel B: Results for market model regressions with control variables						
		Market Liquidity		DGSVI		
	Number of observations	Coefficient	t-statistic	Coefficient	t-statistic	adjusted.R ² (%) F-value
Volume	49,958	0.987***	25.20	0.125***	7.81	20.54% 179.42***
Amihud	49,958	0.998***	26.73	0.093***	4.54	20.84% 180.35***
CS	49,958	1.001***	32.59	−0.013	−0.69	5.82% 194.86***
RHL	49,958	0.996***	37.81	0.088***	7.47	31.65% 316.13***
HLR	49,958	0.996***	37.05	0.084***	7.49	32.08% 353.43***
VOV	49,958	1.002***	35.74	0.082***	7.38	32.06% 340.00***
Panel C: Robustness tests (Chordia, Roll, and Subrahmanyam 2000) for the sign and significance of market liquidity measures						
	Without control [Equation (1)]		With control [Equation (2)]			
	Positive and significant	Positive and insignificant	Positive and significant	Positive and insignificant		
Volume	50	2	50	3		
Amihud	51	2	51	2		
CS	52	1	52	1		
RHL	52	1	52	1		
HLR	52	1	52	1		
VOV	52	1	52	1		

*, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

Panels A and B summarize the results of market model regressions in the panel fixed-effect framework [Equations (1) and (2)]. Results include the regression coefficients and their t-statistics for the market liquidity (and DGSVI measure), *adjusted. R²*, F-statistics, and the number of observations. For brevity, the table does not report the statistics pertaining to control variables. DGSVI $[= \ln(\frac{GSVI_t}{GSVI_{t-1}})]$ measure reported in Panel B is google search volume index (GSVI) corresponding to the 'full name' of currency. The other variant of the GSVI measure corresponding to the symbol of the currency yields similar results (not reported for brevity). Following Chordia, Roll, and Subrahmanyam (2000), the regression models [Equations (1) & (2)] are also estimated separately on the individual CCs (53). In Panel C, we report the number of CCs for which the t-statistics for market liquidity measures (volume, Amihud, CS, RHL, HLR, and VOV) are positive and significant, and positive but not significant.

the CCs. Moreover, the majority of market participants in these cryptocurrencies share common traits; these are tech-savvy committed retail investors that broadly fall under the definition of noise-traders (influenced by behavioural biases such as overconfidence and the representative heuristics; Barber, Odean, and Zhu 2008). We argue that a set of similar factors (e.g. news reports, attention-grabbing public events) attract the attention of these retail investors, and in turn, contribute to liquidity commonality.

V. Conclusion

This paper investigates the phenomenon of liquidity commonality in the cryptocurrency (CC) market, using 53 CCs. The results show that CCs exhibit higher commonality compared to other asset classes. The quantile regression approach to commonality suggests

that the behaviour of commonality is significantly different around extreme quantiles than that observed during normal conditions. We contend that the unique transaction cost structure integral to the CC market – increasing volumes beyond certain thresholds driving up the transaction costs – attenuates the escalating commonality around extreme liquidity quantiles. Moreover, investor attention – measured using GSVI – appears to play an important role in driving the liquidity commonality in CCs. The results suggest that the rising investor attention relates positively to the spread measures (i.e. illiquidity). The observed relationship between the liquidity and investor attention is in contrast to that observed for other asset classes, and may be ascribed to the unique transaction cost structure of the CC market. Portfolio managers are particularly concerned about the liquidity commonality behaviour around extreme quantiles, as the rising

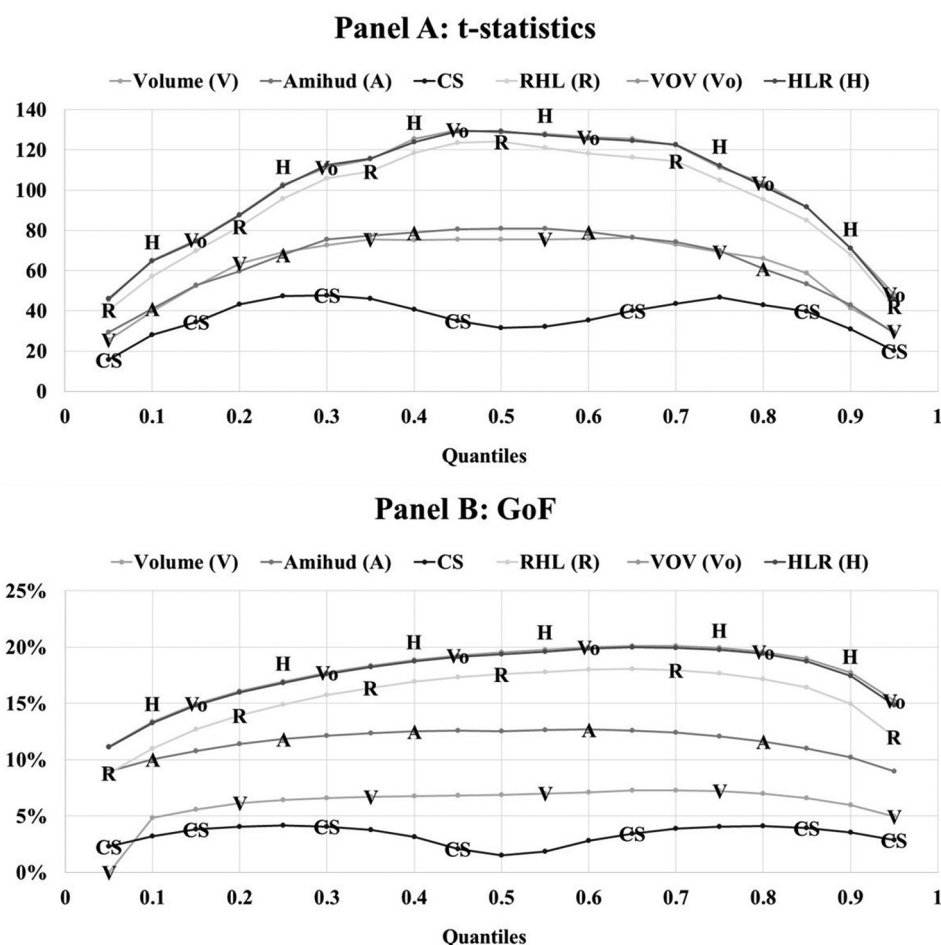


Figure 1. Quantile regression results [Equation (1)] **across the conditional quantiles of liquidity.** **Note:** The figure summarizes the results from the model [Equation (1)] in a quantile regression framework, across the conditional quantiles of liquidity. The time series of the daily changes in the aggregate (market) liquidity of CCs (contemporaneous, lead, and lag) act as the independent variables. The corresponding values of the daily changes in the liquidity of individual CCs act as the dependent variable. Panels A & B show the t-statistics and goodness-of-fit measure (GoF), respectively. The results are shown for the volume, Amihud, CS, RHL, HLR, and VOV measures. The horizontal axis represents 20 quantiles with increments of 0.05. The goodness-of-fit measure is calculated as one minus the ratio of the appropriately weighted sum of absolute errors (deviations) from the conditional model and that from the unconditional model. The goodness-of-fit measure in the quantile regression framework is provided by the 'goodfit' function in the package 'WRTDStidal'. In nature, this function is similar to the conventional R^2 measure for regression. Detailed results corresponding to this figure are provided in Table B1 of Appendix B.

transaction costs can offset the diversification benefits. Therefore, the results from this study are useful to investors as well as academics interested in alternative asset classes such as CCs.

Confirmation of corresponding author

As Corresponding Author, I Abhinava Tripathi, regarding the paper submitted under the title 'Liquidity commonality in the cryptocurrency market', declare that the manuscript is original, has not been published before and is not currently being considered for publication elsewhere. I can confirm that the

manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. I further confirm that the order of authors listed in the manuscript has been approved by all of us. I understand that the Corresponding Author is the sole contact for the Editorial process and is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

Declaration of interest

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Appendix A

Table A1: Literature survey on liquidity commonality.

Study	Data, variables, and methodology	Evidence
Chordia, Roll, and Subrahmanyam (CRS, 2000)	1169 stocks listed on NYSE are examined for a period of 254 days. Measures: QS, PQSPR, Depth, ES, PESPR. Method: Market model of CRS.	The individual firm-specific regressions report low levels of commonality ($\text{adj.}R^2$ in the range of 1–4%). However, the observed levels of commonality increase significantly in the portfolio regressions, as the firm-specific effects are offset ($\text{adj.}R^2$ in the range of 15%–80%).
Brockman and Chung (2002)	All the 725 firms listed on SEHK, over the sample period from 1996 to 1999 (903 trading days). Measures: RS, AS, volume, depth. Method: Market model of CRS	The paper documents the presence of commonality in the Hong-Kong market, a pure order-driven market, devoid of designated market makers.
Pástor and Stambaugh (2003)	Stocks listed on NYSE/AMEX over the period 1966–1999. Measures: Monthly liquidity measure constructed from daily return volume data. Method: Fama-French four-factor model.	Marketwide aggregate liquidity is a priced factor. Smaller stocks are less liquid and have high sensitivity to marketwide liquidity.
Coughenour and Saad (2004)	470 stocks traded on NYSE are examined over the 19-month period ending December 2000. Measure: ES, PES, QS, PQS. Method: Market model of CRS	The study finds that individual stock liquidity co-varies with the specialist portfolio liquidity ($\text{adj.}R^2$ of 22%). This co-variation increases with the risk of liquidity provision. Overall, the evidence suggests that commonality in liquidity is generated by common market makers.
Chordia, Sarkar, and Subrahmanyam (2004)	Treasury bond market (GovPX Inc.) and equity market data (NYSE) is examined over the period 1991–1998. Measures: Time-weighted average quoted spread, depth, OIB. Method: Vector auto-regressions (VAR) model.	The study finds evidence to suggest that common factors drive liquidity and volatility in these markets. The study also examines the linkage between the macro (Fund-flow) and micro (transactions) liquidity.
Domowitz, Hansch, and Wang (2005)	Stocks traded on ASX-20 for 211 days (March 2000 to December 2000). Measure: Round-trip transaction cost, supply and demand schedules of the limit-order book. Method: Simulation, correlations. OLS regressions.	Liquidity commonality is driven by supply and demand of liquidity (i.e. correlation in market and limit order flows). In contrast, return commonality is driven by order direction (buy vs. sell).
Galariotis and Giouvis (2007)	FTSE100 securities are examined over the period 1996–2001. Measures: AS, RS. Method: Market model of CRS.	The study examines the presence of systematic marketwide liquidity for the U.K. market, in the context of changing microstructure from quote-driven to order-driven.
Korajczyk and Sadka (2008)	4055 NYSE listed stocks are examined for the period 1983 to 2000. Measures: Amihud (2002) measure, Turnover (ratio of monthly traded volume and shares outstanding), QS, ES, impact cost measures (λ, Ψ). Method: Factor decompositions, Asymptotic principal components (APC) and Expectation maximization (EM)	They find the presence of a common liquidity factor, particularly in the spread measures and the fixed component of the impact cost. Almost 50% of the variation in individual firm spreads is explained by the first three principal components. This systematic factor in liquidity exhibits persistence, with highly autocorrelated lagged terms. Lastly, they also find that across-measure of systematic risk carries a significant premium.
Kamara, Lou, and Sadka (2008)	Stocks listed on NYSE/AMEX (73,933 firm-year observations) over the period 1963–2005, are examined. Measures: Amihud (2002) Method: Market model of CRS.	The paper shows that the behaviour of systematic liquidity vis-à-vis small and large stocks has evolved differently. This divergence is ascribed to the increasing institutional ownership in the U.S. market.
Kempf and Mayston (2008)	30 stocks from DAX30 (listed on Frankfurt Stock Exchange- Germany) are examined for a 3-month period 2004. Measures: AS, RS, and depth. Method: Market model of CRS.	The study examines liquidity commonality beyond the best prices, capturing the systematic component of liquidity in large trades.
Hameed, Kang, and Viswanathan (2010)	1800 Stocks listed on NYSE are examined for the period 1988 to 2003. Measures: PQSPR, PESPR. Method: OLS time-series regressions.	The paper shows that commonality increases during the period of market declines. They also document that commonality is positively related to market volatility but unrelated to idiosyncratic volatility. Overall, the evidence suggests the presence of a contagion in liquidity due to supply-side effects.
Comerton-Forde et al. (2010)	37 specialists firm from NYSE (and the stocks assigned to them) are examined over the 11-year period 1994–2004. Measures: Percentage spread, Value weightage effective spread, Revenues from a roundtrip transaction, revenues from inventory holding. Method: Time-series OLS regressions.	The study shows that market makers' balance sheet and income statements explain the time-variation in the liquidity, indicating the important role of supplier-financing constraints. This individual specialist position also plays an important role in determining marketwide liquidity.
Cao and Wei (2010)	1589 stock options traded on the U.S. stocks for the period of 1996 to 2004. Measure: Volume weighted average proportional bid-ask spread, price impact, contract volume, dollar trading volume, Amihud measure. Method: Market model of CRS.	The article substantiates the finding that more informed traders tend to trade in options market. The commonality in option market is higher for small and more volatile stocks. Option market liquidity is linked to the underlying stock market liquidity. Moreover, the option market liquidity responds asymmetrically to marketwide movements. Lastly, call and put behave differently to same marketwide movements.
Corwin and Lipson (2011)	100 NYSE stocks are examined over the study period from November 1997 to February 1998. Measures: QS, limit spread, 1000-share spread. Method: Principal Component Analysis	The relative contribution of institutional traders, retail traders, and exchange members is examined in driving commonality in order flow, returns, and liquidity. Programme traders and institutional investors are primary drivers of commonality in returns, order flows, and liquidity.

(Continued)

Table A1: (Continued).

Study	Data, variables, and methodology	Evidence
Marshall, Nguyen, and Visaltanachoti (2013)	Sixteen commodities from agricultural, energy, industrial metal, precious metal, and livestock commodity markets (part of S&P Goldman Sachs Commodity index [S&P GSCI]) are examined over the period of 1997–2009. Measures: Amihud (2002). Method: (a) Market model of CRS.(b) Principal Component Analysis approach.	They find a significant presence of systematic liquidity factor in the commodities market. They find modest evidence to support both the supply- and demand-side factors affecting liquidity commonality. Energy commodities provide the best liquidity risk hedge against the stock market liquidity. Liquidity commonality is higher following market declines. Liquidity withdrawals followed by large price declines provide support to demand-side explanation to liquidity commonality. Higher commonality associated with fund-ownership provides support to demand-side explanations.
Kang and Zhang (2013)	1000 stocks listed on NYSE are examined for one-year period (2003). Measure examined: Limit-order book dispersion (LD). Cost to trade (CT), Proportional quoted spread. Method: Market model of CRS.	The liquidity measures computed for the outside (deeper) and inside (top) of a limit-order book behave differently. Overall, the study provides evidence of commonality in the liquidity provided by limit-order book. In the market model regressions, the inside liquidity exhibits an average $\text{adj.}R^2$ of 31.64%, and the outside liquidity exhibits an average $\text{adj.}R^2$ of 6.80%
Mancini, Rinaldo, and Wrampelmeyer (2013)	Nine currency pairs (AUD/USD, EUR/CHF, EUR/GBP, EUR/JPY, EUR/USD, GBP/USD, USD/CAD, USD/CHF, and USD/JPY exchange rates), obtained from Electronic Broking Services (EBS), are examined over the period of 2007–09. Measures: Price impact measure of Kyle (1985), PQSPR; price dispersion measure. Method: Principal component analysis,	The study presents the evidence of commonality in the foreign exchange (FX) market. The study shows that there are large co-movements in the liquidity of FX market. The evidence suggests that more liquid securities tend to have lower sensitivities to marketwide FX liquidity. The liquidity in the equity, bond, and FX markets co-moves more strongly when the markets are illiquid.
Rösch and Kaserer (2013)	160 companies listed on the major German market indices (DAX, MDAX, SDAX, TecDAX), traded on Xetra market (German market) are examined over the study period from 2003 to 2009. Measures: Volume weighted average spread measure. Method: Market model of CRS.	The study documents that commonality varies over time, increases during market downturn, peaks around crisis events, and becomes weaker around the deeper levels of order book. Funding liquidity plays an important role in driving this observed liquidity commonality.
Frino, Mollica, and Zhou (2014)	Nine index futures markets are examined over the 10-year period 2002–2012. Measures: QS, RS, and market depth. Method: Brockman et al. (2002) model is employed.	The study provides significant evidence of the presence of global liquidity commonality in the index futures market. Their evidence suggests that commonality in liquidity has increased over time.
Koch, Ruenzi, and Starks (2016)	Stocks listed on NYSE/AMEX over the period of 1980 to 2010 (121,592 stock-quarters) are examined. Measures: Amihud (2002). Method: Market model of CRS.	Highly correlated trading activity among the stocks, held by mutual funds, drives commonality and provides the demand-side explanation to the phenomenon of commonality. The effect of mutual fund ownership is particularly strong during the outflows.
Moshirian et al. (2017)	39 markets over a period of 15 years (1996–2010) are examined. Measures: RS. Method: Market model of CRS.	The study shows that liquidity commonality is driven by both market-level and firm-level factors. Overall, the evidence suggests that liquidity commonality is higher in volatile environments, with poor investor protection, and high information asymmetry.

Abbreviations

Stock exchanges: New York Stock Exchange (NYSE), American Stock Exchange (AMEX), Deutscher Aktienindex (DAX, Germany), Financial Times Stock Exchange (FTSE), Stock exchange of Hong-Kong (SEHK), Australian Securities Exchange (ASX).

Measures: Quoted Spread (QS), Proportional Quoted Spread (PQSPR), Depth (DEP), Effective spread (ES), Proportional effective spread (PESPR), Relative bid-ask spread (RS), Absolute spread (AS), Percentage effective spread (PES), Percentage quoted spread (PQS), Order imbalance (OIB).

Chordia, Roll, Subrahmanyam (CRS, 2000).

Appendix B

Table B1: Quantile regression results [Equation (1)] across the conditional quantiles of liquidity.

Panel A: Volume measure										
Quantiles ($\tau = .05$ to 0.50)										
Variables	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t</i> -stat (Market)	25.53	39.77	52.47	63.60	69.05	72.65	75.50	75.28	75.38	75.44
GoF	3.94%	4.86%	5.61%	6.16%	6.44%	6.61%	6.70%	6.76%	6.83%	6.91%
Quantiles ($\tau = .55$ to 0.95)										
Variables	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t</i> -stat (Market)	75.62	75.77	76.48	73.13	69.25	65.89	58.64	41.31	29.79	
GoF	7.01%	7.13%	7.26%	7.29%	7.23%	7.01%	6.59%	5.96%	4.95%	
Panel B: Amihud measure										
Quantiles ($\tau = .05$ to 0.50)										
Variables	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t</i> -stat (Market)	29.36	41.17	52.80	59.70	67.96	75.38	77.30	79.04	80.56	81.01
GoF	8.97%	10.06%	10.80%	11.38%	11.85%	12.15%	12.38%	12.53%	12.60%	12.56%
Quantiles ($\tau = .55$ to 0.95)										
Variables	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t</i> -stat (Market)	81.05	79.35	76.64	74.30	69.79	61.11	53.39	43.03	29.11	
GoF	12.66%	12.69%	12.60%	12.40%	12.09%	11.64%	11.02%	10.22%	8.97%	
Panel C: CS measure										
Quantiles ($\tau = .05$ to 0.50)										
Variables	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t</i> -stat (Market)	15.64	28.14	34.56	43.31	47.23	47.85	46.02	40.62	35.06	31.70
GoF	2.29%	3.19%	3.82%	4.06%	4.18%	4.08%	3.76%	3.13%	2.10%	1.54%
Quantiles ($\tau = .55$ to 0.95)										
Variables	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t</i> -stat (Market)	32.10	35.34	40.06	43.71	46.85	42.91	39.93	30.80	20.07	
GoF	1.87%	2.83%	3.46%	3.87%	4.08%	4.13%	3.94%	3.53%	2.87%	
Panel D: RHL measure										
Quantiles ($\tau = .05$ to 0.50)										
Variables	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t</i> -stat (Market)	40.42	57.10	69.79	81.93	95.83	105.89	109.34	118.42	123.46	124.11
GoF	8.82%	11.03%	12.70%	13.94%	14.93%	15.74%	16.39%	16.94%	17.33%	17.59%
Quantiles ($\tau = .55$ to 0.95)										
Variables	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t</i> -stat (Market)	121.05	118.33	116.21	114.50	104.98	95.31	84.88	67.85	42.22	
GoF	17.79%	17.99%	18.05%	17.95%	17.65%	17.19%	16.42%	14.96%	12.03%	
Panel E: HLR measure										
Quantiles ($\tau = .05$ to 0.50)										
Variables	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t</i> -stat (Market)	46.08	64.64	74.68	87.42	102.15	112.50	115.67	124.09	129.23	129.30
GoF	11.13%	13.26%	14.84%	15.96%	16.84%	17.62%	18.23%	18.75%	19.13%	19.38%
Quantiles ($\tau = .55$ to 0.95)										
Variables	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t</i> -stat (Market)	127.56	125.72	124.72	122.73	112.20	102.17	91.76	71.15	45.88	
GoF	19.62%	19.86%	19.99%	19.96%	19.77%	19.40%	18.74%	17.46%	14.85%	
Panel F: VOV measure										
Quantiles ($\tau = .05$ to 0.50)										

(Continued)

Table B1: (Continued).

Panel A: Volume measure										
Quantiles ($\tau = .05$ to 0.50)										
<i>Variables</i>	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t-stat</i> (Market)	45.39	65.13	75.32	87.96	102.64	111.34	115.46	125.49	130.02	128.71
GoF	11.20%	13.38%	14.98%	16.08%	16.96%	17.73%	18.34%	18.86%	19.25%	19.51%
Quantiles ($\tau = .55$ to 0.95)										
<i>Variables</i>	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	
Number of observations	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958	49,958
<i>t-stat</i> (Market)	127.94	126.36	125.46	122.34	111.38	103.57	91.67	71.02	48.42	
GoF	19.76%	20.00%	20.11%	20.11%	19.94%	19.59%	18.98%	17.75%	15.23%	

The table presents the results from the model [Equation (1)] in a quantile regression framework, across the conditional quantiles of liquidity. The time series of the daily changes in the aggregate (market) liquidity of CCs (contemporaneous, lead, and lag) act as the independent variables. The corresponding values of the daily changes in the liquidity of individual CCs act as the dependent variable. Panels A-F show the results corresponding to volume, Amihud, CS, RHL, HLR, and VOV measures. The results show the number of observations, t-statistic for the contemporaneous market liquidity changes, and Goodness-of-fit (GoF) measure for the regressions.

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